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Innovative Deployment of Floating Solar PV: A Case Study of RWH and STF

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Abstract

This paper presents a pioneering case study on the successful design, implementation, and performance of Floating Solar Photovoltaic (FSPV) systems ("float-o-voltaic") at Bharat Petroleum Corporation Ltd. (BPCL)'s Kochi Refinery (KR), Kerala. Installed on two artificial water bodies - the Rainwater Harvesting (RWH) pond and the Shore Tank Farm (STF) - this initiative marks India's first floating solar project executed by an Oil & Gas Public Sector Undertaking (PSU). The project's primary objectives include augmenting the share of green electricity in refinery operations, optimizing unused water surface areas, reducing Scope-2 emissions, and aligning with BPCL's corporate goal of achieving net-zero emissions by 2040. A detailed internal feasibility study was conducted by the Renewable Energy Business Unit (RE BU) in collaboration with the KR engineering team to evaluate the solar potential of various available non-utilized spaces within the refinery, including ground and water bodies. While a ground-mounted 8.4 MWp (DC)/6 MW (AC) solar PV system was finalized for the 25.83-acre land bank, floating solar projects were concurrently explored over the RWH and STF ponds. These locations were selected owing to their consistent water availability, proximity to the internal power distribution system. The floating solar projects were commissioned in early 2024 and integrated seamlessly into the KR internal power grid. The systems are generating green power at an LCOE of approximately ₹3.5/kWh, demonstrating high economic viability. Performance data shows a capacity utilization factor of ~24%, attributed to the natural cooling effect of water enhancing panel efficiency. Additionally, the installations contribute to water conservation by reducing surface evaporation by an estimated 60 kL annually per MW installed. The FSPV installations have led to significant reductions in grid power dependency (12 MW average import previously) and have realized substantial monetary savings, while contributing to sustainability metrics under ESG reporting.

This project sets a benchmark for innovation in the Indian O&G sector by demonstrating the technical and commercial feasibility of floating solar systems in industrial complexes. The strategic use of idle water bodies as energy generating assets minimizes land-use conflict and paves the way for replicable models across BPCL's refineries and depots nationwide. The learnings from this deployment offer a template for scalable, safe, and sustainable solar integration within high energy demand industrial settings.

INTRODUCTION

India's energy scenario

As per the latest figures, India's total primary energy supply stands at 903,158 Kilotonnes of Oil Equivalent (KToE), growing by 7.8% from the previous period. The per-capita consumption of energy has grown from 14,682 MJ/person in 2014-15 to 18,410 MJ/person in 2023-24, resulting in a CAGR of 2.55%. Electrical energy forms the backbone of industrialization and energy transfer in any economy and in India, close to 1,800 billion units (kWh) of energy was generated in the last financial year. With India's large population of over 1.45 billion people and a main consumption-driven economy, the power demand in the country is expected to grow by 15 GW/year for the next 6 years. Currently, the installed capacity of 475 GW comprises of 50.5% from conventional sources and the rest from renewable and non-conventional sources. (MoSPI 2025)

India's energy transition

In February 2025, India achieved the historic milestone of 100 GW of installed solar power capacity. This feat marks India's commitment to a cleaner, greener future by just and sustainable energy transition through renewable energy sources. The *Panchamrit* ("Five nectar") goals announced by the Government of India in CoP26, Glasgow, United Kingdom, includes reaching 500 GW of non-fossil fuel capacity and fulfilling 50% of the national energy requirement from renewable energy by the end of the decade. Thus, renewable energy sources, including solar, wind, hydro, and biomass energy sources, are crucial for India to achieve its aspirational and ambitious goals in energy transition. These goals also align towards the Sustainable Development Goals of 2030 (SDG), of which Affordable and Clean Energy (SDG-7) seeks to ensure access to affordable, reliance, sustainable modern energy for all. (MNRE 2025)

Solar energy remains the backbone and the dominant contributor to India's renewable energy growth, accounting to nearly 47% of the total installed capacity of renewable energy in the country. India's solar power sector has witnessed an extraordinary 3450 % increase in capacity over the past decade, rising from 2.82 GW in 2014 to 100 GW in 2025. Additionally, India has also made great strides in solar manufacturing. Over the past decade, India's module production capacity has surged to 60 GW (in 2024) from 2 GW. With continuous policy emphasis, India is on track to achieve a solar module production capacity of 100 GW by 2030. Production-Linked incentives for solar modules and import duties have worked in tandem to improve *Atmanirbharta* ("self-reliance") in the solar energy sector.

In this scenario of dynamic developments in the national renewable energy space and solar energy, Bharat Petroleum Corporation Limited (BPCL), a company traditionally in the Oil & Gas Sector, had announced the goal to embark up on a mission to reach Net-Zero status by 2040. The floating solar PV projects taken in this paper for study, are a small step in this giant leap of energy transition.

FLOATING SOLAR PV: AN INNOVATIVE RENEWABLE BREAKTHROUGH

Genesis of the floatovoltaic

The origin of the concept of floating solar PV (FPV) emerged in the early part of the new millennium. The increasing need for innovative renewable energy solutions to replace traditional technologies which had reached their maturity and the scarcity of land, caused by rapid urbanization, growth in agriculture and industrial development, forced countries like Japan and France to look towards alternatives. It is in this light that FPVs were born. Japan's first FPV plant was installed in 2007, followed by France with small-scale pilot projects. The enhanced efficiency due to water-based cooling and reduced water evaporation and algal growth, made FPVs an attractive alternative proposition for green energy transition.

Global adoption and growth

The global FPV market was valued at USD 7.7 billion in 2024 and is projected to grow at a CAGR of 1.5% from 2025 to 2034. This shows strong fundamentals and a bright outlook in the market economy for FPVs. The major push for FPV growth is concentrated along three broad regions – Southeast Asia, Europe, West Asia & Africa. Rapid adoption in countries like China, Vietnam and Indonesia are enhanced by the robust manufacturing and workforce capabilities of these economies. In Europe, the Netherlands and France are taking the lead in FPVs, with the West Asia & Africa using the technology to combat water scarcity and high temperatures.

Arrival and evolution of Floating Solar in India

India's entry into the FPV sphere began in 2014-15, with pilot installations in Kerala and West Bengal. With creation of a dedicated Ministry (Ministry of New and Renewable Energy – MNRE) and collaboration with international agencies like GIZ (Germany), a Whole-of-Economy and Whole-of-Government approach was harnessed to launch India's RE potential. This policy push is evident throughout the renewables space, with rapid growth in wind, bioenergy and other forms of non-conventional energy technologies. Notable FPV projects in India include the 101.6 MWp FPV plant executed by Tata Power in Kayamkulam, Kerala and the 100 MW FPV at Ramagundam, Telangana.

Future Potential

A massive potential of 206.7 GWp from FPVs have been evaluated under a report prepared under the Indo-German Technical Cooperation on Innovative Solar, a quantity close to the total installed capacity of Brazil. Over one-third of this total potential is concentrated in the states of Madhya Pradesh and Maharashtra. The FPV industry is quite literally a sunrise sector, with a cumulative capacity of 30 GW expected by 2040. With improvements in efficiency with R&D, the cost of setting up an FPV plant is expected to reduce by 2.5% annually and the Levelized Cost of Energy (LCOE) would possibly be reduced by 33% in the same timeframe.

Globally, expansionist trends for offshore FPVs are also being observed (Netherlands etc.) along with integration in hydropower and desalination plants. However, it is crucial to note that environmental and ecological concerns, related to impact of aquatic life, along with technical challenges involved in anchoring and maintenance are to be addressed effectively for efficient and sustainable use of the technology.

A TECHNOLOGICAL OVERVIEW OF FLOATING SOLAR PV TECHNOLOGY

OVERVIEW

Floating Solar PV (FPV), also known as "floatovoltaics" involves installing solar panels on water bodies like reservoirs, ponds, lakes and dams. It is a land-neutral solution ideal for areas with land scarcity or high land costs. This approach not only conserves valuable land resources but also enhances energy efficiency through natural cooling provided by the water surface. Thus, floatovoltaics offer a high-potential, land-neutral and technically robust solution for meeting industry needs, particularly in areas with water bodies which can be harnessed for the same. Their adoption combines innovation with environmental stewardship.

The PV panels, being installed on floating water bodies such as dams and ponds, eliminate the requirement of land availability, foundations and Module Mounting Structures (MMS) compared to ground mounted systems. The anchored floats would be attached to the shores of reservoir bottoms as required to avoid movement during wind, wave etc. These floats also come with a fixed tilt arrangement. Additionally, PV modules have a negative temperature coefficient and hence will generate more energy as compared to land mounted systems due to cooling effect of the water.

SYSTEM COMPONENTS AND CONFIGURATION

PV Modules – Types

Silicon-based solar PV modules are the dominant type of modules available in the market and widely used. However, technological advancements have results in manufacturing PV modules from thin-film technologies also. Presently, the most commonly used technologies are mono-crystalline and TOPCon. Crystalline silicon technology dominates the global installed capacity of solar PV and accounts for approximately 95% of the total market.

1. *Monocrystalline Passivated Emitter and Rear Cells (Mono - PERC)*

The Mono-PERC modules use monocrystalline silicon cells (made from a single crystal) with passivation layer added on the rear side of the cells. This layer reflects unabsorbed sunlight back into the cell, giving the light a second chance to be absorbed, thus increasing the cell's efficiency. They also perform better in low-light and high-temperature conditions.

Due to their high efficiency and performance under varying conditions, they have become the go-to choice for solar module production. The global PERC solar market was valued at \$140.4 billion in 2022 and is projected to reach \$304.9 billion by 2032, with a CAGR of 8.2%. Domestically, mono PERC modules accounted for 85.3% of solar module supplies to India in 2024.

2. *Polycrystalline PERC*

These are similar to the Mono-PERC technology, however they differ in the fact that they are constructed using polycrystalline silicon cells, made by melting and fusing multiple silicon crystals together. Though, less efficient than Mono-PERC, Poly-PERCs are popular due to their lower cost.

The global polycrystalline solar cell market was valued at \$3.53 billion in 2024 and is expected to reach \$7.01 billion by 2034. Domestically, they accounted for only 0.9% of the total solar module supplies to India in 2024.

3. *Tunnel Oxide Passivated Contact (TOPCon)*

The TOPCon technology involves a thin tunneling oxide layer placed between a transparent conductive oxide layer and a p-doped crystalline silicon layer. This structure minimizes charge carrier recombination and significantly boosts efficiency. They tend to exhibit higher efficiency than PERC and perform well at relatively higher temperatures.

TOPCon technology is gaining traction as a high-efficiency alternative to PERC. In India, they have accounted for ~14% of the solar module supplies in 2024. Leading domestic manufacturers are offering TOPCon modules at affordable prices.

4. *Other parameters*

In addition to these parameters, there are also Half-Cut Cells, created by splitting traditional silicon solar cells (either monocrystalline or polycrystalline) into two or more smaller pieces using a non-destructive technique. This reduces resistive losses and increases efficiency. Bifacial solar panels are designed to capture sunlight from both the front and rear sides, increasing energy production. They are especially beneficial in locations with reflective surfaces like snow or water. Double glass modules (also known as Glass-Glass modules) have glass on both the front and back sides, replacing the traditional polymer backsheet. This construction offers improved durability, resistance to temperature fluctuations, and is particularly relevant for bifacial designs, allowing for light absorption from the rear

Selection of Solar PV Technology for FSPV Plants. Considering the maturity, long-term proven track record and availability of the crystalline silicon technology, crystalline silicon technology are mostly preferred for FSPV installations. TOPCon modules are also being explored for possible applications. Beyond performance of PV modules, failure modes should be considered when selecting PV modules for use in FSPV plants. FSPV plants encounter higher risks of exposure to moisture, mechanical stresses, and potential induced degradation (PID).

Given the above circumstances and technology status, the following aspects should be considered in selecting PV modules for FSPV installations:

1. **Preferred Technology:** Use advanced modules like **mono PERC** or **mono PERC half-cut cells** for higher efficiency.
2. **Glass-Glass Modules:** Offer better sealing but are heavier, requiring stronger floatation and mounting. Frameless designs may be less rigid and prone to moisture ingress if not sealed properly.
3. **Encapsulation:** **Olefin-based encapsulants** are recommended for better moisture and UV resistance.
4. **Mechanical Stress:** **Half-cut cells** and **multi-bus bar designs** help reduce strain and maintain performance even with minor cell damage.
5. **PID Resistance:** Most manufacturers now offer **PID-resistant modules**, which should be a standard requirement.
6. **Anti-Soiling Coatings:** Special coatings reduce dust accumulation and shading losses, lowering maintenance needs.

Floats. Depending on the type of application, different types of floats are available in the market:

1. **Module Pontoon:** These are primarily made of High Density Polyethylene (HDPE) components and require anchoring either to the reservoir banks or under water. The anchoring design depends on geotechnical survey, bank suitability and the inspection & maintenance costs. Various anchoring methods are available including steel anchors with shackles and cables, concrete block placed underwater and bank or pole-based anchoring (which is cheaper and easier for Operation & Maintenance).
2. **Modular Raft:** These structures use a rigid metal frame placed on a floating base. These rafts have the advantage of easy sourcing and transport along with the added rigidity which reduces the stress on connections in the PV module. However, the rigidity may contribute to increased mechanical stress at weak points due to water movement.
3. **Ferrocement float:** These floats are constructed utilizing thin-wall reinforced concrete and wire mesh. These structures have the added advantage of crack control, impact resistance and durability, offering toughness due to uniform reinforcement dispersion.

Floating membrane. The membrane is an impermeable system made of polymer placed on water. It is secured by a floating HDPE circular pipe. The PV modules are mounted on aluminum rails fixed with keder tracks. These membranes can withstand high wind speeds and the direct water contact can help reduce the module operating temperature. However, the rainwater accumulation required pumping and special modules (double-sided and frameless) are required to harness the complete potential of the FPV system. They also may suffer problems due to soiling issues.

Novel FPV technologies.

1. **Submerged FPV:** The PV modules in the system are placed just below the water surface. This ensures enhanced cooling from direct water contact and reduced mechanical stress from the wind and waves. This is an experimental technology and is in nascent stages of deployment at small-scale projects.
2. **High-Wave Offshore FPV:** This involves utilizing vast ocean space near coastal cities. However, due to intense waves, wind, tides, currents and the corrosive marine environment, this involves a whole host of challenge. Currently, it is at a conceptual level and requires significant technological advancement.

Selection of floats. The majority of large-scale FPV plants deployed worldwide have been carried out using HDPE floats where the solar PV modules are installed on pontoons above the water. However, recent technological advancements have resulted in floating platforms with Basic K2 Systems float (with pontoons and metal frames) which offer higher air circulation to the modules. Both these float technologies have enough manufacturing base and can be manufactured locally.

Anchoring systems. To fix the PV system on the water, various different anchoring systems can be considered:

1. **Fixed Anchoring on shore:** The fixed anchoring system secures floating solar PV (FSPV) islands to the shore using submerged cables, with minimal visual impact. It accounts for water level variations and uses a chain connection at the island end. The system is robust, non-intrusive to the environment, and allows for easy maintenance without requiring divers. Anchors are installed on both the shoreline and the FSPV island, ensuring stability and long-term performance in varying aquatic conditions.
2. **Submerged anchoring system:** The submerged anchoring system uses deadweight anchors placed near the shoreline to secure floating solar PV (FSPV) installations. This method avoids ground penetration, making it suitable for sensitive or protected environments. However, it requires a larger footprint due to the size and placement of the weights. The system provides stable anchorage while minimizing ecological disruption, offering a viable alternative to pile-based anchoring methods for FSPV projects.
3. **Submerged anchoring system fixed to the bottom:** This anchoring method is used when the floating solar PV (FSPV) platform is located far from the shore. It involves securing the platform directly to the waterbed using submerged anchors. This system ensures stability in deep or remote water bodies without relying on shoreline structures. It is particularly suitable for large reservoirs or lakes where shore-based anchoring is impractical. The design provides robust support while minimizing surface-level interference and visual impact.

Mooring systems. Mooring systems for floating solar platforms can be passive or active, with passive systems preferred due to lower complexity and cost. Three main passive configurations are used: **catenary**, **multi-catenary**, and **taut**. Catenary systems rely on line-weight and friction, while taut systems use line tension and require anchors that withstand uplift forces. Multi-catenary systems combine materials for flexibility and reduced environmental impact. Hybrid systems with intermediate buoys are ideal for fluctuating water levels, offering stability and adaptability.

1. **Catenary Mooring:** This system uses heavy mooring lines laid parallel to the lakebed. The restoring force comes from the line's weight and friction with the soil. It is simple to install but not suitable for vertical loads or floaters with limited connection strength. It requires space and has a larger environmental footprint.
2. **Multi-Catenary Mooring:** Combines different materials in the mooring line, allowing restoring forces from both weight and deformation. It suits platforms with many connection points and limited load capacity. Intermediate buoys can help manage vertical movement. It is more complex to install but has a reduced environmental impact.
3. **Taut Mooring:** Uses tensioned lines anchored at an angle to the lakebed. Restoring forces come from line deformation. It is simple to install but requires stretchable lines and anchors that can handle significant uplift forces. It is ideal for areas with minimal vertical water displacement and has a low environmental footprint.

Mooring lines for floating solar platforms are classified into chain, wire, and synthetic types, often used in hybrid combinations. Chains provide weight and friction at the seabed, while synthetics act as stretchers.

Selection involves collaboration among manufacturers, selectors, and developers. Steel wire ropes and High Modulus Synthetic Fibers (HSMF) are stiff and suitable for minimal movement, whereas synthetics are preferred for flexibility. Key performance factors include abrasion, fatigue, creep, UV degradation, and corrosion. Among synthetics, polyester is most durable, nylon offers high stretch but loses strength when wet, and polypropylene/polyethylene have lower abrasion resistance and applicability.

Inverter. Inverters convert DC electricity from solar panels to AC for grid use and optimize energy output. They are classified into micro, string, and central inverters. String inverters suit small installations with shading or orientation issues but require many units for utility-scale projects. Central inverters are cost-effective for large plants and can be housed on land or floating platforms. Module-based MPPT systems offer high-resolution monitoring and safety but come with higher CAPEX. The selection of inverters depends on plant size, layout, and the performance needs required.

String Inverters

String inverters are smaller devices designed to connect a limited number of PV modules. They are particularly effective in optimizing electrical parameters and maximizing energy production in installations where module orientation or shading varies, such as rooftop systems. However, for utility-scale solar plants, the need for a large number of string inverters can lead to increased installation and maintenance costs. These costs include not only the initial setup but also the potential need for replacements over the plant's operational lifetime.

Central Inverters

Central inverters are high-capacity devices, typically rated at 1,000 kW or more, and are used to manage larger sections of a solar plant. While they may not optimize energy production as efficiently as string inverters, they are generally more economical to purchase and install, and less resource-intensive to maintain. For utility-scale installations, central inverters are usually housed in weatherproof, skid-mounted enclosures similar in size to shipping containers. Alternatively, they can be installed in masonry buildings with proper ventilation. In some cases, central inverters may be placed on floating platforms within specialized containers, provided they meet a minimum ingress protection (IP) rating of 65. This floating setup is considered when the reduction in electrical cable losses outweighs the benefits of land-based accessibility.

Module-Based MPPT Systems

Some manufacturers offer inverter systems that include a DC-DC power converter and a Maximum Power Point Tracking (MPPT) device on each PV module or pair of modules. These systems enable high-resolution monitoring of individual solar panels, allowing for early detection of performance issues or technical failures. They also minimize losses due to panel mismatch or soiling, as each MPPT device independently controls its connected module's output without affecting the entire string. Additionally, these systems enhance safety for operations and maintenance personnel by limiting DC string voltages to 30 V when no heartbeat signal is received from the inverter. Although this technology involves higher capital expenditure, the potential for improved energy yields may justify the cost, which can be assessed during project feasibility studies.

Cabling. Cabling in FPV systems involves DC cables connecting modules to inverters and AC cables routing power to transformers and the grid. Additional cables include communication lines for SCADA integration and grounding cables for safety compliance. Two main configurations are used: **floating cables**, which are economical and easy to maintain but visually intrusive, and **submerged cables**, which offer low visual impact but are costlier and harder to install. Floating cables are often preferred for their convenience and affordability.

Floating Cables

Floating cables are enclosed in waterproof sheaths and secured with fasteners, allowing them to float on the water surface. This method is economical, easy to install, and simple to maintain, making it a popular

choice for FSPV projects. However, it has a visual impact and may obstruct boat movement, which can be a concern in navigable water bodies.

Submerged Cables

Submerged cables are routed underwater in impermeable sheaths from the floating island to the shoreline. This configuration offers a low visual impact, making it suitable for aesthetically sensitive or high-traffic areas. However, it is more expensive and challenging to install and maintain, requiring specialized equipment and procedures.

Transformers. Transformers in PV systems are essential for adjusting voltage levels. They either step up medium-voltage AC from inverters to high-voltage levels (6.6 kV and above) for grid connection or step-down voltage for lower requirements. Selection must comply with IS/IEC/IEEE standards. Utility-scale PV projects commonly use three-phase oil-type transformers. Often, integrated units combining inverters, transformers, and switchgear are specified, capable of stepping up voltage to 6.6, 11, 22, or 33 kV. These units simplify installation and enhance system efficiency.

Switchgear. Switchgear in PV systems manages the connection between the solar plant and the electricity distribution network. It must comply with IS/IEC/IEEE standards, with separate switchgear units for the plant and the local network operator, typically housed in distinct cabins. The switchgear should be indoor, metal-clad, floor-mounted, and fully draw-out type, built from 2.0 mm thick cold-rolled steel. It must support extension at either end and be compartmentalized for safety and maintenance. Protection levels should be IP42 for indoor and IP65 for outdoor installations. Standard engineering practices require flush-mounted relays, meters, switches, and lamps, along with necessary control and interlock systems. Mechanical interlocks are recommended in addition to electrical ones to ensure safe operation and switching procedures.

SCADA System. Weather stations play a crucial role in FSPV plants by collecting real-time environmental data such as solar radiation, module and ambient temperature, humidity, wind speed and direction, rainfall, wave activity, water temperature, and more. This data is essential for evaluating plant performance, detecting underperformance, and identifying opportunities for design or maintenance improvements. The weather station is typically installed on the floating island and integrates with a centralized monitoring system.

This monitoring system aggregates data from the weather station along with inputs from modules, inverters, transformers, and plant meters. It forms part of the plant's SCADA (Supervisory Control and Data Acquisition) system, which enables remote access, control, and operation of key equipment. The SCADA system, along with the Power Plant Controller (PPC), includes a human-machine interface (HMI), input/output modules, communication infrastructure, and software. Industry-standard practices recommend using an industrial Ethernet LAN and fiber optic network distributed across the plant to ensure reliable communication with field device

BACKGROUND & SITE OVERVIEW

Background

Kochi Refinery has an internal electricity demand of approximately 160 MW, which is currently met through a combination of its captive power plant (CPP) and imports from the Kerala State Electricity Board (KSEB) grid, averaging around 12 MW. To support the transition to green energy, the Renewable Energy Business Unit (RE BU) and Kochi Refinery (KR) teams conducted a joint survey to identify unused land within the refinery premises suitable for solar power development. Based on this survey, the team identified several large water bodies within the refinery and at the Shore Tank Farm (STF) area in Puthuvype as potential sites for floating solar installations. Furthermore, representatives from RE BU along with experts in the domain, visited these water bodies to assess their suitability for solar deployment. The team also visited existing

solar installations for reference, including the 427.7 kW DC / 360 kW AC floating solar plant at the Cochin International Airport Golf Club and CIAL's 14.4 MW DC / 12 MW AC ground-mounted solar facility.

These site visits and assessments are part of a broader initiative to integrate renewable energy into Kochi Refinery's operations, reduce grid dependency, and promote sustainable energy practices within the refinery ecosystem. (BPCL 2023)

Site Overview

Rainwater Harvesting Pond Floating solar power project at Kochi Refinery. Kochi Refinery, India's largest public sector refinery, operates with an installed crude refining capacity of 15.5 MMTPA and an internal electricity demand of approximately 164 MW. This demand is primarily met through its Captive Power Plant (CPP), supplemented by a 220 kV grid connection from Kerala State Electricity Board (KSEB). Additionally, the refinery procures electricity from the open market via power exchanges.

As part of its renewable energy integration strategy, Kochi Refinery has identified two rainwater harvesting ponds as potential sites for FPV installations. Pond-1 spans 12 acres and Pond-2 covers 6.83 acres. These water bodies can accommodate a floating solar plant with a capacity of 4.6 MW DC / 3.2 MW AC. Based on PVSyst simulations, the estimated annual energy generation is 6,853 MWh. Considering the thermal advantage of floating systems, which operate at lower temperatures than ground-mounted modules, a conservative 1.5% increase in generation is projected, resulting in an annual output of approximately 6,956 MWh.

Additionally, the absence of space constraints near the pond allowed for convenient placement of Balance of System (BOS) components and cable routing.

Floating Solar Project at Shore Tank Farm. The Shore Tank Farm (STF) at Kochi Refinery, located approximately 20 km from the main refinery complex at Puthuvype, serves as the crude receipt facility. Unlike the main refinery, STF does not have a captive power plant and relies entirely on imported electricity from the KSEB.

STF contains four fire water ponds—two large and two small—covering a total area of 13.96 acres. These ponds have been identified as suitable sites for deploying an FPV system. Preliminary assessments indicated the potential to install a 5.2 MW DC / 3.7 MW AC floating solar plant. Based on PVSyst simulations, the estimated annual energy generation is 7,667 MWh. Given the thermal advantage of floating solar systems, which operate at lower temperatures than ground-mounted modules, a conservative 1.5% increase in generation was projected, resulting in an annual output of approximately 7,781 MWh.

The electricity generated was evacuated through substation at STF, located about 1 km from the proposed site. With no space constraints near the pond, the Balance of System (BOS) components has been installed adjacent to the solar array, and connectivity established.

PROJECT LIFE CYCLE

Initiation

The solar projects at both the locations were done through Public-Private Partnership (PPP) mode of project execution. The Engineering, Procurement and Construction (EPC) contractor and the Project Management Consultant (PMC) were selected through an open tender with competitive bidding to minimize the cost for maximum returns. The PMC undertook the basic designing of the FPV project, with layouts and engineering drawings made for the floats, electrical systems and power evacuation. A comprehensive Single Line Diagram (SLD) of the entire electrical system at both the locations was also prepared. All designs made by the PMC were comprehensively evaluated by a two – step process; both at KR and RE BU. This allowed for multiple checks both on-site as well as off-site, along with cross-functional inputs. An additional benefit of this process was technical knowledge was shared and strengthened for the teams both at HQ (RE BU)

and on Field (KR). Through multiple revisions all finalized drawings were set into motion in the project pipeline. An overall project timeline and schedule with continuous monitoring was also envisaged, to ensure accountability and smooth flow of communication. The approved drawings were then discussed with the EPC contractor, whose role being critical in the execution stage, shared critical feedback on the on-ground feasibility of certain designs (float placement, module stringing etc.). These were amicably discussed with the PMC and with effective co-ordination from RE BU and KR, the on-ground execution started.

Execution

Being the first FPV project in India at an oil installation, the project execution involved challenges which often stalled the project for short period of time. Preparation of the pond in itself proved to be a daunting task with significant silting on the pond beds. With efforts from the EPC contractor, de-silting was carried out at both the sites. Thick algal growth on the surface was also cleared using manual methods. The laying of the floaters was a critical node in the project execution network. The factory-inspected floats had to undergo various tests including wind force withstanding ability. Accordingly, the floats were installed at the ponds. The electrical systems were installed at the banks of the ponds along with the control room for the overall monitoring of the station. Due to the shadowing effect near the banks, a few branches of trees were also trimmed. The entire electrical system includes transformer, inverters and cabling as mentioned earlier. Through systematic efforts by the site team and continuous monitoring on the delivery of items, the team was able to successfully complete the project.

Commissioning

The Pre-Commissioning activities were started gradually as the components were installed and tested. After rigorous checking and third-party inspections, the site team was able to do all the necessary safety checks and inter-connection with the internal grid of the Refinery. Accordingly, the charging and commissioning of both the sites were carried out successfully and the plants were formally inaugurated by the Chief Secretary, Government of Kerala.

SITE PICTURES



Fig. 1a



Fig. 2b



Fig. 3c



Fig. 4d

Figure 1a-d—Floating Solar Project at RWH and STF, Kochi Refinery, Kerala (pictures taken at site)

GENERATION DATA

The yearly snapshot the generation from each of the plants is as given in the table below:

Table 1—Generation Data at the locations (internal report)

Sl.No.	Site Name	Location	Annual Generation (MWh)
1	5.2 MWp DC / 3.8 MW AC Floating Solar Plant at Shore Tank Farm in Kochi	KR	6548.30
2	4.6 MWp DC / 3.4 MW AC Floating Solar Plant at Kochi	KR	5792.73

CONCLUSIONS AND RECOMMENDATIONS

The successful deployment of the FPV systems at BPCL's Kochi Refinery marks a significant milestone in India's renewable energy landscape, particularly within the Oil & Gas sector. By utilizing unconventional water bodies, namely the Rainwater Harvesting (RWH) pond and Shore Tank Farm (STF), this initiative not only optimized idle infrastructure but also demonstrated the technical and economic viability of floating solar technology in industrial settings. With a capacity utilization factor of ~24% and a competitive levelized cost of energy (LCOE) of ₹3.5/kWh, the project has achieved substantial reductions in grid power dependency and Scope-2 emissions, while contributing to water conservation and ESG performance metrics.

This pioneering effort aligns with BPCL's strategic vision of achieving net-zero emissions by 2040 and developing a 10 GW renewable energy portfolio by 2035. The learnings from this deployment offer a

replicable and scalable model for future installations across BPCL's refineries and depots nationwide. As industrial energy demands continue to rise, the integration of FPV systems presents a forward-looking solution that balances sustainability, safety, and operational efficiency, setting a benchmark for innovation in renewable energy adoption within high-demand industrial environments.

The project's success underscores the viability of FPV systems as scalable and replicable models for energy generation in constrained environments. It also highlights the importance of interdisciplinary collaboration in engineering, environmental planning, and energy management to achieve sustainable outcomes. The learnings from anchoring, mooring, and electrical integration have set new benchmarks for future deployments.

In conclusion, these floating solar projects exemplify how innovative deployment strategies can accelerate the adoption of renewable energy in industrial settings. They serve as a testament to the potential of floating solar technology in supporting India's and the world's clean energy transition, offering a blueprint for similar initiatives globally.

NOMENCLATURE

AC	Alternating Current
BOS	Balance of System
BPCL	Bharat Petroleum Corporation Limited
CAGR	Compound Annual Growth Rate
CPP	Captive Power Plant
DC	Direct Current
ESG	Environmental, Social, and Governance
EPC	Engineering, Procurement, and Construction
FSPV / FPV	Floating Solar Photovoltaic
GW	Gigawatt
HDPE	High-Density Polyethylene
HMI	Human-Machine Interface
IEC	International Electrotechnical Commission
IP Rating	Ingress Protection Rating
KSEB	Kerala State Electricity Board
KToE	Kilotonnes of Oil Equivalent
KR	Kochi Refinery
kWh / MWh	Kilowatt-hour / Megawatt-hour
LCOE	Levelized Cost of Energy
MNRE	Ministry of New and Renewable Energy
MMS	Module Mounting Structure
MPPT	Maximum Power Point Tracking
MW / MWp	Megawatt / Megawatt-peak
O&M	Operation and Maintenance
PID	Potential Induced Degradation
PMC	Project Management Consultant
PPC	Power Plant Controller
PSU	Public Sector Undertaking
PV	Photovoltaic
RE BU	Renewable Energy Business Unit
RWH	Rainwater Harvesting
SCADA	Supervisory Control and Data Acquisition

SDG Sustainable Development Goals
SLD Single Line Diagram
STF Shore Tank Farm
TOPCon Tunnel Oxide Passivated Contact
UV Ultraviolet

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